Continoun Rondom Variable

the Join function Rondom variable x and y

1) $P(x = a, y = y) = f(x, y)$

2) $f(x, y) \geq 0$

3) $\int_{\alpha}^{\alpha} \int_{\alpha}^{a} f(x, y)$ change

the Join function x and y in this case ie

$f(u.v) = P(x = x, y = y)) = \int \int f(u.v)$ (2)

$u = -a \; v = -a$

by differen the distrubit function setting by x and y

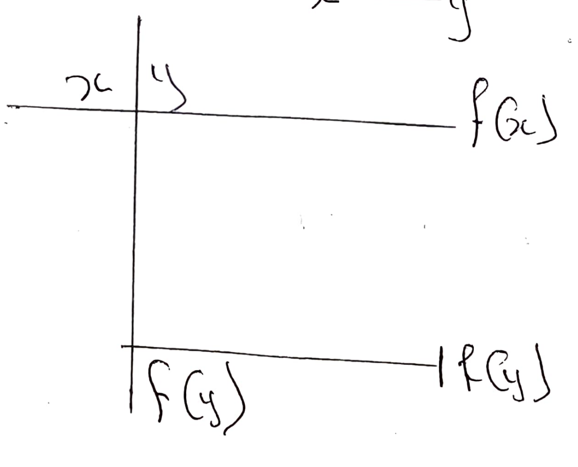
$P(x = x)_2 \; f(x) = \int \int_{u = -a} f(u.v)$ — (3)

$P(y = y) = f(y) = \int^{x} \int^{y}_{-\alpha \; r = \alpha} f(u.v)$ abedy (4)

$f(x, y) \geq 0$

$\int_{-\alpha}^{\alpha} \int_{\alpha}^{\alpha} (x, y) y = 1$

$\sum_{x} \sum_{y} f(x, y) = 1$



$f(x,y) = c\,(2x+y)$ $0 \le x < 2 : 0 \le y \le 3$

What function Probability :- Is the resource of a Join function to a masure Probability.

Find the value of C.

① C

② $P(x=2, y=1)$

③ $P(x \ge 1, y \le 2)$

Solve

$f(x,y) = c\,(2x+y)$

$c = (2\cdot 0+0) = 0$

$c = (2\cdot 0+1) = 1$

$c = (0+1) = c$

$c(2x+y)$

$c(2\cdot 1+0)$

$c(2\times 1+1)$

$c(2\cdot 2+0)$

$c(2\cdot 2+1)$ $5c$

$x \backslash y$	0	1	2	3	$f(x)$
0	0	c	2c	3c	6c
1	2c	3c	4c	5c	14c
2	4c	5c	6c	7c	22c
$f(y)$	6c	9c	12c	15c	$42c = \sum_x \sum_y f(x,y)$

Since grande total is $42c$ and be equal 1

(i) $42 = 1$

$$c = \frac{1}{42}$$

(ii) $P(x=2, y=1) = 5c = \frac{5}{42}$

(iii) $(2c+3c+4c) + (4c+5c+6c)$

$$\sum_{x \geq 1} \sum_{y \leq 2} f(x,y) = (2c+3c+4c) + (4c+5c+6c)$$

$$= 24c = \frac{24}{42}$$

find the marginal function of $f(x,y) =$

Solve

$f\ c = \frac{1}{42}$ $\quad 6c = \frac{6}{42}$

$14c = \frac{14}{42}$

$22 = \frac{22}{42}$

$f(x) = 6c = \frac{6}{42}$

$9c = \frac{9}{42}$

$12 = \frac{12}{42}$

$f(y) = 15c = \frac{15}{42}$

Other Example

the Join densting function of 2 Continoun Random Variable x and y is

$$f(x,y) = \begin{cases} cxy : 0 < x < 4 ; 1 < y < 5 \\ 0 : \text{otherwasi} \end{cases}$$

find the Variable

1. Variable c
2. $P(1 < x < 2 ; 2 < y < 3)$
3. $P(2x \geq 3, y \leq 2)$

Solve

1. $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y) = 1$

$= \int_0^4 \int_1^5 cxy = 1$

$= c \int_0^4 x \frac{y^2}{2} \Big|_1^5 = 1$

$= \int_0^4 x \left(\frac{25}{2} - \frac{1}{2}\right) dx = 1$

$= \int_0^4 12x\,dx = 1$

$12c \int_0^4 x = 1$

$12c \left(\frac{x^2}{2}\right)_0^4$

$12 = c\left(\frac{16}{2}\right) = 1$

$= 12c(8) = 1$

$96c = 1$

$c = \frac{1}{96}$

(2) $P(1<x<2; 2<y<3)$

$$= \int^2 \int^3 \frac{xy}{96} \, dx$$

$$= \frac{1}{96} \int_1^2 \frac{xy^2}{2} \Big|_2^3$$

$$= \frac{1}{96} \int_1^2 \left(\frac{9}{2} - \frac{4}{2}\right) dx$$

$$= \frac{1}{96} \int_1^2 \frac{5\cancel{x}}{2} \, dx$$

$$= \frac{5}{192} \int_1^2 = dx$$

$$= \frac{5}{192} \left(\frac{x^2}{2}\right)^2$$

$$= \frac{5}{192} \left(\frac{4}{2} - \frac{1}{2}\right)$$

$$= \frac{5}{\cancel{192}_{64}} \left(\frac{\cancel{3}}{2}\right) = \frac{5}{128}$$

$\therefore$ final $P(1<x<2, 2<y<3) = \frac{5}{128}$

(3) $P(x \geq 3, y \leq 2)$

$$= \int_3^4 \int_1^4 \frac{xy}{96}\, dx$$

$$= \frac{1}{96} \int_3^4 \frac{xy^2}{2} \Big|_1^2\, dx$$

$$= \frac{1}{96} \int_3^4 x\left(\frac{4}{2} - \frac{1}{2}\right) dx$$

$$= \frac{1}{96} \int_3^4 \frac{3x}{2}\, dx$$

$$= \frac{3}{192} \int_3^2 x\, dx$$

$$= \frac{3}{192} \left(\frac{x^2}{2}\right)_3^4$$

$$= \frac{3}{192} \left(\frac{16}{2} - \frac{9}{2}\right)$$

$$= \frac{\cancel{3}}{\cancel{192}\,64} \left(\frac{7}{2}\right) = \frac{7}{128}$$

two

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Assignment

1. find the ~~nature~~ margical of x and y

$$f(x,y) = \begin{cases} \frac{xy}{96} & 0<x<4, 1<y<5 \\ 0 & \text{otherwise} \end{cases}$$

2. the Join Probability function of 2 variable x and y by give)

$$f(x,y) = \begin{cases} cxy: & x=1,2,3,\ y=1,2,3, \\ 0: & \text{otherwosi} \end{cases}$$

i. find c.
ii. $P(x=2, y=1$
iii. $P(1 \le x \le 2, y \le 2)$
iv. $P(x \ge 2)$
v. $P(x=1)$

Condition Probability

the Join Probability of two - densting random variable x and y is give by

Example 1

$$f(x,y) = \begin{cases} c(2x+y) ; & 0 \le x \le 2 \\ 0 & 0 \le y \le 3, \; 0, 1, 2, 3. \end{cases}$$

find the consdence of

(i) c

(ii) $f(y | x=2)$

(iii) $f(y=1 | x=2)$

Solve

$$\sum_{x=0}^{2} \sum_{y=0}^{3} f(x,y) = 1$$

(i) $c = \frac{1}{42}$

(ii) $f(y | x=2) = \frac{f(x,y)}{f(x)} = \frac{(2x+y)/42}{f(x)}$

$$= \frac{(2x+y)/42}{\frac{4}{42}}$$

$$= \frac{22}{42} \cdot \frac{2x+y}{42}$$

$$\frac{22}{\ldots} = 2x+y \times \frac{42}{\ldots} = \frac{2x+y}{22}$$

(iii) $f(y=1 \mid x=2) = \frac{\angle 1+1}{22} = \frac{5}{22}$

Example (2)

a, is card drom from the Rondom card play card with you or Ace.

Consender the Probability space that has 52 clensting card as equally liks outcome let A. B event of drom ace and let B event and drowing ane doimon we requir. $P(B/A)$ $A=13$ $\heartsuit=13$ $\spadesuit=13$ $\diamond=13$

Solve

$$P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{1}{52}$$

$$= \frac{1}{52} \times \frac{13}{1} = \frac{13}{52} = \frac{1}{4}$$

Example (3)

a ball contain block & what ball a preson select 2 ball without replacement if the probability selecting block & what ball is $\frac{15}{56}$

Pdf Probability densting function

function of random variable let x be a random variable then $y = y(x)$ is also random variable.
Let x be c.r.v. the $y = y(x)$ is also r.v. it follow that $P(y=y) = \sum_{x/g(x)=y} P(x=x)$

Solve

① if x 5 dire hand Probability function $f(x)$

$E\{g(x)\} = g(x) f(x) g(x) f(x_2) g(x) f(x_i)$

$$= \sum_{i=1}^{x} g(x) f(x_2)$$

② if x 5 C.r.v. hand Pdf the $E g(x) \int_{\alpha}^{x} f(x) dx$

Some there are on Expectation.

1) if c is any constant the $E(cx) = cE(x)$

2) if x and y are any r.v the $E[x+y] = E(x) + E(y)$

3) if x and y are Indenpendent r.v the $E[xy]$
$= E(x) \cdot E(y)$

another quantity of Important in Probability and Statatic is the variance are $(x) = \sigma_x^2 = E[(x-a)^2]$

Solve

$= \sqrt{\text{vand}(x)} = \sqrt{E(x-u)^2}$

Examples

the Join densting function of variable x and y by give $f(x,y) = 1$

$$f(x,y) = \begin{cases} \frac{2xy}{96} & 0 < x < 4, 1 < y < 5 \\ 0 & \text{otherwise} \end{cases}$$

find

1. $E(x)$
2. $E(y)$
3. $E(xy)$
4. $E(2x+3y)$

Solve

1) $E(x) = \int_{-\infty}^{4}\int_{\infty}^{5} x\, f(x,y)\, dx\, dy$

$= \int_{0}^{4}\int_{1}^{5} x \frac{2xy}{96}\, dx\, dy$

$= \frac{1}{96}\int_{0}^{4} x^2 \frac{y^2}{2}\Big|_{1}^{5}\, dx$

$= \frac{1}{96}\int_{0}^{4}\left(\frac{25x^2}{2} - \frac{x^2}{2}\right) dx$

$= \frac{1}{96}\int_{0}^{4} x^2\, dx$

$= \frac{1}{96}\int_{0}^{4} 12x^2\, dx$

$= \frac{12}{96}\int_{0}^{4} x^2\, dx$

$= \frac{12}{96}\left[\frac{x^3}{3}\right]_{0}^{4}$

$= \frac{12}{96}\left[\frac{64}{3}\right] = 8/3$

$\therefore E(x) = 8/3$

$$E(y) = \int_0^4 \int_1^5 \frac{y\,xy}{96}\,dx\,dy$$

$$= \int_0^4 \int_1^5 \frac{xy^2}{96}\,dx\,dy$$

$$= \frac{1}{96}\int_0^4 \frac{xy^3}{3}\Big|_1^5\,dx$$

$$= \frac{1}{96}\int_0^4 \left(\frac{125x}{3} - \frac{x}{3}\right)dx$$

$$= \frac{1}{96}\int_0^4 \frac{124}{3}x\,dx$$

$$= \frac{124}{288}\int_0^4 x\,dx$$

$$= \frac{124}{288}\left[\frac{x^2}{2}\right]_0^4$$

$$= \frac{\overset{31}{\cancel{\overset{62}{\cancel{124}}}}}{\underset{9}{\cancel{\underset{18}{\cancel{288}}}}}\left(\frac{16}{2}\right) = \frac{31}{9}$$

$$\therefore E(y) = \frac{31}{9}$$

(3) $E(x,y) = E(x)\,E(y)$

$= \left(\frac{8}{3}\right)\left(\frac{31}{9}\right)$

(4) $E(2x+3y) = E(2x) + E(3y)$

$= 2E(x) + 3E(y)$

$= 2\left(\frac{8}{3}\right) + 3\left(\frac{31}{9}\right)$

Central limet thoarem

let x be a random variable with me σ^2 if $\bar{X}_n$ is the means of the random Sample of Size of x chose from the denstins of x then the denstion.

$$\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}\right)$$

tenst to standar normal densting in as $n \to \infty$ when we draw a large random Sample from The Population who me is new and variable Signment variable Into $\bar{X}_n$

a test us that A is larg. the distribution $\left(\bar{X}_n \frac{-\mu}{\sigma/\sqrt{n}}\right)$ is aprosimall narmal with is 0 1. what is marthorble about the theorem is there the no speace nature of all we need to know about distributs with me and variable.

theorem 1 $<+x_1\ x_2 --- x_2$ be Indenpend random variable which in random variable which are finite means μ and variance σ^2

then if $F_A = X_1 + X_2 + X_3 ---- + X_n$

$$\lim_{n\to\infty} P\left[a \le \frac{S_n - n\mu}{\sigma\sqrt{n}} \le b\right] = \frac{1}{\sqrt{2\pi}} \int_a^b e^{\frac{-x^2}{2}}\, dx$$

$$\frac{1}{\sqrt{2\pi\sigma^2}} \int_a^b e^{-\frac{1}{2}(x-\mu)^2}\, dx$$

that is random variable $\frac{S_n - n\mu}{\sigma\sqrt{n}}$

which std variable S_n is assigntotically normal the theorem is true order more generated function. for example L theorem holder when this variance $X_1\ X_2\ ---X_n$ is inderPeden variable with means.

Example

11/08/2022

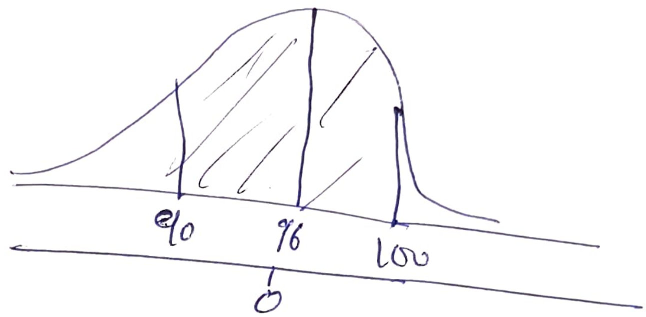
The average age of a vehicle registered in a country is 96 months. Assume that their stated deviation is 16 month. If a random sample of 36 vehicle is selected find the probability that the mean of their ages is b/w 90 and 100 months

Solution

Let x be age

$P(90 \leq x \leq 100$

x_1 x_2



$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

$n = 36$

$z_1 = \frac{90-96}{16\sqrt{36}} = \frac{6}{16/6} = \frac{-6}{2.7} = -2.2$

$z_2 = \frac{100-96}{16/\sqrt{36}} = \frac{4}{2.7} = 1.5$

$P(z_1 < z < z_2) = P(z_1 = -2.2) + P(z_2 = 1.5)$

$P(0.4861 + 0.4332)$

$\therefore P(96 \leq x \leq 100) = 0.91934 \times 100$

$= \underline{91.93\%}$

Example 2

A researcher reported ~~that~~ children b/w age of 2 and 5 watch on average of 25 hours of a cartoon programs on TV per week. Assume the variable is normally distributed and the standard deviation is 3 hours. If 20 children b/w the ages of 2 and 5 are randomly selected, find the probability that the mean of the numbers of hour their watch TV will be ≥ 26.3 hour

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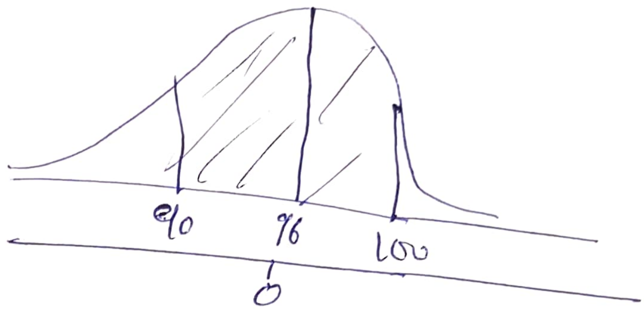
Let x be age

$P(90 \leq x \leq 100)$, with $x_1 = 90$ and $x_2 = 100$



$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$n = 36$

$$Z_1 = \frac{90-96}{16\sqrt{36}} = \frac{6}{16/6} = \frac{-6}{2.7} = -2.2$$

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$$P(z_1 < z < z_2) = P(z_1 = -2.2) + P(z_2 = 1.5)$$

$$P(0.4861 + 0.4332)$$

$$\therefore P(96 \leq x \leq 100) = 0.9193 \times 100$$

$$= \underline{91.93\%}$$

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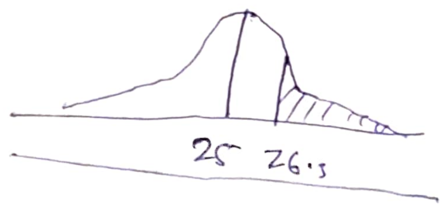
Solution

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$



$$Z = \frac{26.3 - 25}{3/\sqrt{20}} = \frac{1.3}{0.6708} = 1.94$$

$$P(Z > 1.94) = 0.5 - P(Z < 1.94) = 0.5 - 0.4738 = 0.0262$$

NB

$$Z = \frac{X - \mu}{\sigma} \qquad Z = x - \text{new} \not\Sigma \text{ segma (standard deviation)}$$

The first one is used to get information about individual data value when the variable is normally distributed.

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

= is used get information when applying the central limit theorem when the variable is normally distributed or when the sample size grater than 30 or more

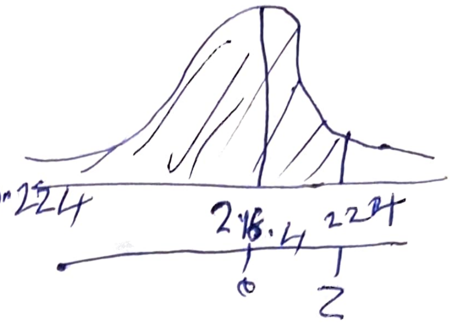
Example 3

The average numbers of Pound of meat a person consume in a year is 218.4 Pounds. Assume that the standard deviation is 25 Pounds and the distribution is approximatly normall.

(a) find the probability that a person selected at random consumes less than 224 Pounds of meat per year

(b) If a sample of 40 individual is selected find the probability that the mean of the sample will be < 224 Ponds per year.

Solution



$$Z = \frac{\bar{x} - m}{\sigma} = \frac{224 - 218.4}{25} = 0.224$$

$$= P(Z < 0.224) = 0.5\cancel{871} + 0871$$

The area b/w 0 and 0.224 is 0.0871
It must be added to 0.5 i.e. 0.5 + 0.0871

$P(x < 224) = \underline{0.5871}$

Estimate of Population is called Parametre
" " Sample is " Statisti